AKSHAR AI

Summer Internship Report

Chest X-Ray Pneumonia Detection System

Deep Learning Based Medical Diagnosis

Name	Jainam Pandya
Enrollment No.	24162121019
Branch	B.Tech Computer Science & Engineering
Institute	Ganpat University
Organization	Akshar AI Ltd
Report Period	Week 1
Project	Pneumonia Detection System
GitHub Repository	`https://github.com/2006jainampandya-cyber/Intership-work`

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Contents

1. Introduction

The Chest X-Ray Pneumonia Detection System is a deep learning based medical image analysis project aimed at detecting pneumonia from chest radiographs.

During Week 1, the primary focus was on dataset acquisition, dataset exploration, exploratory data analysis (EDA), GitHub repository management, and frontend development using React.js.

The dataset was analyzed using Kaggle Notebook and Python libraries. A simple React homepage was also developed to provide the initial user interface for the project.

2. Weekly Objectives

- Download and understand the Chest X-Ray Pneumonia dataset.
- Verify dataset structure and image organization.
- Perform Dataset Loading and Understanding.
- Perform Exploratory Data Analysis (EDA).
- Study image characteristics and class imbalance.
- Develop a React homepage.
- Manage project source code using GitHub.

3. Weekly Progress Summary

Task	Status
Dataset Setup	Completed
Dataset Verification	Completed
Class Distribution Analysis	Completed
Image Visualization	Completed
Resolution Analysis	Completed
Pixel Intensity Analysis	Completed
Artifact Inspection	Completed
React Homepage Development	Completed
GitHub Repository Setup	Completed

Table 1: Week 1 Progress Summary

4. Day-wise Work Summary

Day 1

- Project orientation.
- Understanding project workflow.
- Environment setup.

Day 2

- Downloaded Chest X-Ray Pneumonia dataset.
- Configured Kaggle Notebook.
- Explored dataset hierarchy.

Day 3

- Verified train, test and validation folders.
- Counted images across categories.
- Completed dataset understanding.

Day 4

- Generated sample image visualizations.
- Performed class distribution analysis.
- Conducted resolution analysis.
- Performed pixel intensity analysis.
- Conducted artifact inspection.

Day 5

- Developed React homepage.
- Created navigation and hero section.
- Added feature cards and footer.
- Uploaded source code to GitHub.

5. Methodology

Dataset Loading

Python libraries such as OpenCV, PIL, and OS were used to load and validate the dataset.

Exploratory Data Analysis

EDA was performed using Matplotlib and Seaborn to study image distributions, class imbalance, image dimensions, and artifacts.

Frontend Development

React.js was used to create a simple homepage containing navigation, hero section, features section, and footer.

6. Tools and Technologies Used

Technology	Purpose
Python	Data Analysis
NumPy	Numerical Computing
Pandas	Data Processing
Matplotlib	Visualization
Seaborn	Statistical Visualization
OpenCV	Image Processing
PIL	Image Handling
React.js	Frontend Development
GitHub	Version Control
Kaggle Notebook	Development Platform

7. Results and Observations

Dataset Distribution

Dataset Split	NORMAL	PNEUMONIA
Train	1341	3875
Test	234	390
Validation	8	8

Table 2: Dataset Distribution

Important Findings

- Dataset is heavily imbalanced towards the Pneumonia class.
- Image dimensions vary significantly.
- Pixel intensity distributions differ between classes.
- Structural artifacts such as R/L markers are present.
- React homepage prototype was successfully created.
- GitHub repository was established for version control.

8. Screenshots

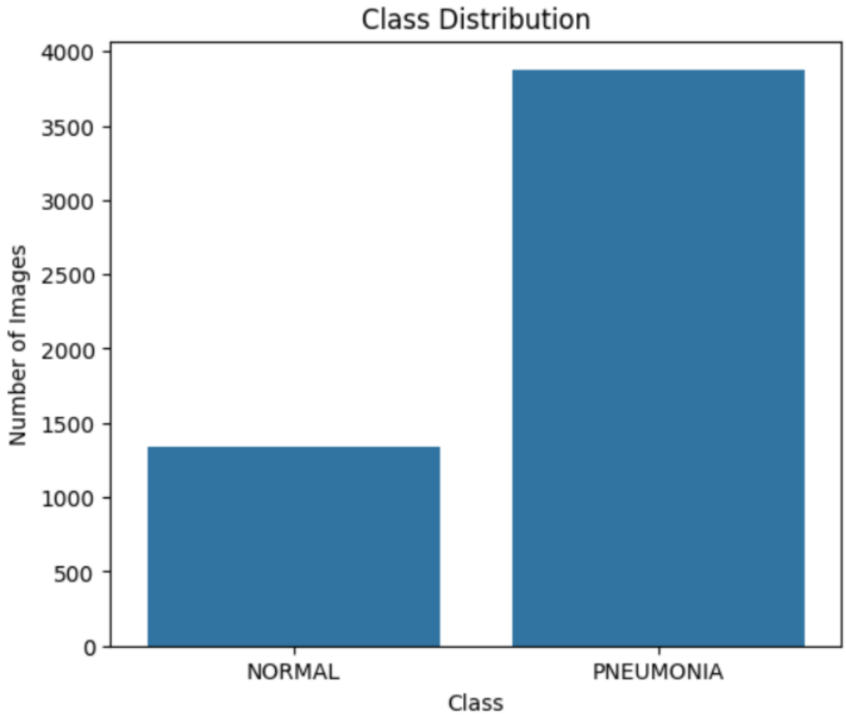

Figure 1: Class Distribution Analysis

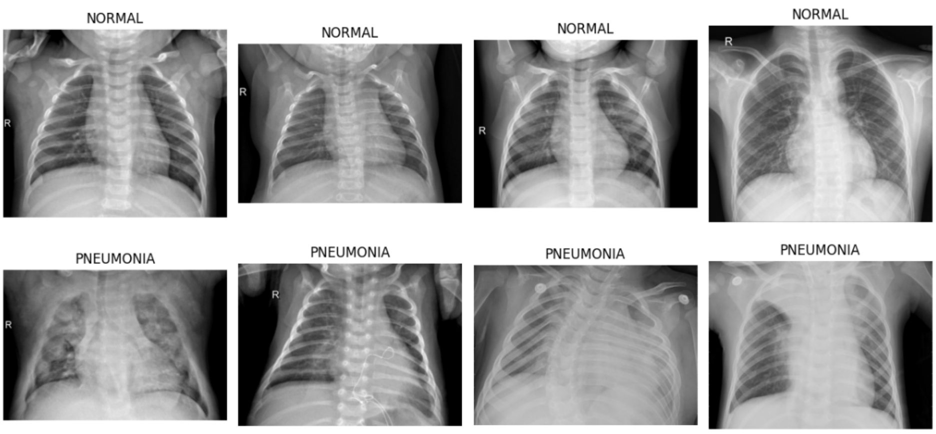

Figure 2: Sample X-Ray Images

Figure 3: Resolution Analysis

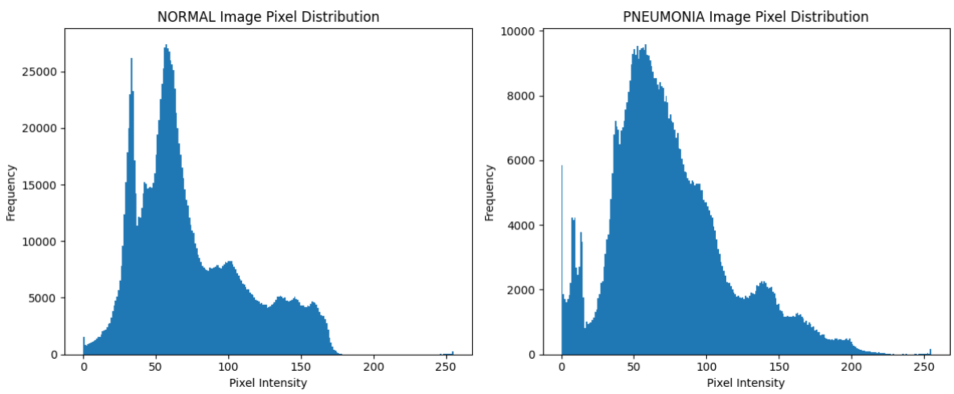

Figure 4: Pixel Intensity Distribution

9. Challenges Faced

- Understanding nested dataset structure.
- Managing class imbalance.
- Handling varying image resolutions.
- Identifying structural artifacts.
- Designing a clean frontend interface.

10. Learning Outcomes

- Learned medical image dataset exploration.
- Improved Python EDA skills.
- Gained practical experience with Kaggle.
- Learned image visualization techniques.
- Improved React.js development skills.
- Enhanced GitHub workflow understanding.

11. Future Work

The next phase of the project will focus on:

- Image preprocessing.
- Data augmentation.
- CNN model development.
- Model training and validation.
- Performance evaluation.
- Deployment of prediction system.

12. Conclusion

Week 1 established a strong foundation for the Chest X-Ray Pneumonia Detection System project. The dataset was successfully loaded, verified, and analyzed through multiple EDA techniques. Important observations regarding class imbalance, image variability, and structural artifacts were identified.

A React homepage was also developed and the project repository was maintained through GitHub. The work completed during Week 1 provides a strong base for preprocessing, augmentation, model training, and deployment activities planned in future weeks.

Prepared By:
Jainam Pandya